Chapter	Title	Total periods	Total time	Date
Ch 05	Prime Time	10	460 min	23 May 2026

C No.	Targeted competencies
C-1.2	Discovers, identifies, and explores patterns in numbers and describes rules for their formation (e.g., multiples of 7, powers of 3, prime numbers), and explains relations between different sets of numbers (whole numbers, fractions, integers, rational numbers, and real numbers)
C-1.4	Explores and understands sets of numbers, such as whole numbers, fractions, integers, rational numbers, and real numbers, and their properties, and visualises them on the number line
C-6.1	Applies both inductive and deductive logic to formulate definitions and conjectures, evaluate and produce convincing arguments/proofs to turn these definitions and conjectures into theorems or correct statements, particularly in the areas of algebra, elementary number theory, and geometry
C-7.1	Demonstrates creativity in discovering one's own solutions to puzzles and other problems, and appreciates the work of others in finding their own, possibly different, solutions
C-9.2	Knows and appreciates the contributions of specific Indian mathematicians (such as Baudhayana, Pingala, Aryabhata, Brahmagupta, Virahanka, Bhaskara, and Ramanujan)

Period 1 — 40 min — Idli-Vada Circle Game: Discovering Common Multiples

Material	Open floor space for circle game, Textbook page 107-108, Board/chart for Venn diagram sketch — **Pedagogical approach:** Play-way
0-8	Introduce and play Activity (Idli-Vada Game), section 5.1 p.107 ("students replace multiples of 3/'idli', 5/'vada', both/'idli-vada'...."). Run two full rounds so students experience eliminations and the 'idli-vada' moments naturally.
8-18	Pause the game and ask: when does everyone say 'idli-vada' at the same time? Guide a brief class discussion to surface the idea of common multiples. Use Fig. 5.1 from the textbook to organise multiples of 3 and 5 into overlapping Venn regions on the board.
18-32	Students work individually on Figure it Out Q1, section 5.1 p.108 ("find the 10th common multiple of 3 and 5") and Figure it Out Q2, section 5.1 p.108 ("how many times 'idli'/'vada'/'idli-vada' said up to 90"). Circulate to check counting strategies and address off-by-one errors.
32-40	Class discusses Figure it Out Q4, section 5.1 p.108 ("draw Venn diagram of multiples of 3 and 5 up to 60"). Preview next period: the same game logic works in reverse to find common factors.
Homework	Figure it Out Q3, section 5.1 p.108 — What if the game was played till 900? How would the answers to Q2 change?
Teacher Notes	This is the opening period; briefly remind students what multiples and factors mean from prior learning before launching the game. Watch for the common error of including 0 as a multiple when counting 'idli-vada'. Students who finish early may begin Figure it Out Q3, section 5.1 p.108.

Period 2 — 40 min — Jump Jackpot: Using Common Factors to Reach Both Treasures

Material	Textbook pages 109-111, Number line drawn on board, Pair-work worksheets (optional) — **Pedagogical approach:** Problem-solving
0-10	Recap common multiples from Period 1, then introduce Activity (Jump Jackpot), section 5.1 p.109 ("Jumpy selects jump size; can only land on multiples of that size...."). Pose the problem: if treasures are on 12 and 18, which jump sizes reach both? Let students guess before computing.
10-22	Students work in pairs on Figure it Out Q1, section 5.1 p.111 ("treasures on 28 and 70; find all valid jump sizes") and Figure it Out Q4, section 5.1 p.111 ("find common factors of four number pairs"). Pairs compare answers and discuss how common factors relate to the jump mechanic.
22-33	Work through Figure it Out Q1, section 5.1 p.110 ("find all multiples of 40 between 310 and 410") as a bridge exercise connecting divisibility and the number line. Discuss the key insight: common factors of two numbers are the factors of their GCF.

33-40	Consolidate by asking students to state in their own words the difference between 'common multiple' and 'common factor'. Set Figure it Out Q8, section 5.1 p.111 as homework and preview the next topic: what makes some numbers special - having exactly two factors.
Homework	Figure it Out Q8, section 5.1 p.111 — Guna erased numbers from a Venn diagram leaving only the common multiples 24, 48, 72.…
Teacher Notes	Transition from Period 1: connect the LCM concept (first 'idli-vada') to GCF (largest valid jump size) so students see both ideas as two sides of the same coin. A common error is listing a jump size that divides only one treasure number; use the 28/70 problem to expose this explicitly. Example 2 on the Jump Jackpot page may be used as a self-study pointer for stronger students.

Period 3 — 40 min — Rectangular Arrays and the Sieve: Who Are the Primes?

Material	1-100 number grid (one per student), Board grid for Sieve demonstration, Textbook pages 112-115	**Pedagogical approach:** Inductive

0-8	Narrate the Guna (12 figs, many rectangles) vs. Anshu (7 figs, one rectangle) story from section 5.2. Ask students to try arranging 11, 12, 13, 15 objects into rectangles, then observe which numbers allow only one arrangement - inductively surfacing the concept of a prime number.
8-22	Walk through the Sieve of Eratosthenes step by step on a 1-100 grid on the board: circle 2, strike multiples; circle 3, strike multiples; continue. Students simultaneously work on their own grids; the Math Talk prompt ('why does this procedure work?') is posed mid-activity for a 2-minute pair discussion.
22-33	Students answer Figure it Out Q1, section 5.2 p.114 ("is there any even prime other than 2?"), Figure it Out Q4, section 5.2 p.114 ("which of 23, 51, 37, 26 are prime?"), and Figure it Out Q9, section 5.2 p.114 ("true/false statements about primes and composites"). Review answers as a class, clarifying that 1 is neither prime nor composite.
33-40	Introduce twin primes via Figure it Out Q8, section 5.2 p.114 ("find all twin prime pairs between 1 and 100") as a short exploration using the completed sieve. Assign Figure it Out Q7, section 5.2 p.114 as homework.
Homework	Figure it Out Q7, section 5.2 p.114 — Find seven consecutive composite numbers between 1 and 100.
Teacher Notes	Transition: recap GCF/LCM from the previous two periods and note that understanding factors deeply requires knowing which numbers are prime. The most common error is treating 1 as prime; address this explicitly when defining primes as having exactly two factors. Students may self-study Figure it Out Q10, section 5.2 p.115 on products of exactly three distinct primes.

Period 4 — 60 min — Pattern Hunting Among Primes: Twin Primes and Deeper Properties

Material	Completed 1-100 sieve grids from Period 3, Textbook pages 113-115, Small-group recording sheets	**Pedagogical approach:** Discovery

0-10	Recap the sieve from Period 3 and pose a reasoning challenge: do all decades (1-10, 11-20, ...) contain the same number of primes? Students examine their completed sieves and record decade-by-decade counts for Figure it Out Q3, section 5.2 p.114 ("which decades have fewest/most primes?").
10-25	Small groups work on Figure it Out Q5, section 5.2 p.114 ("prime pairs less than 20 whose sum is a multiple of 5") and Figure it Out Q6, section 5.2 p.114 ("find digit-reversal prime pairs up to 100"). Groups share findings; teacher highlights the reasoning pattern - what makes a reversed-digit number composite most of the time?
25-40	Class works through Figure it Out Q10, section 5.2 p.115 ("which numbers are products of exactly three distinct primes?") together, modelling systematic prime-factor checking. Then students individually attempt Figure it Out Q12, section 5.2 p.115 ("find primes p where 2p+1 is also prime"), recording at least five examples.
40-52	Class discussion: what patterns did they notice in Q12? Can they conjecture whether this always holds? Introduce the historical note on Eratosthenes and the sidebar about the largest known prime to situate prime discovery in a broader context.
52-60	Consolidate reasoning goals: students write one sentence explaining why 2 is the only even prime. Assign Figure it Out Q11, section 5.2 p.115 as homework and preview the next section: what does it mean for two numbers to share no common factor at all?

Homework	Figure it Out Q11, section 5.2 p.115 — How many three-digit prime numbers can you make using each of 2, 4 and 5…
Teacher Notes	Transition: connect the sieve's pattern of 'striking out multiples' to the reasoning goal - primes are precisely the numbers not struck out by any smaller prime. Watch for students reasoning that large odd numbers must be prime; use Q10's composite odd numbers (45, 91, 105) to counter this. Students who finish early may self-study Figure it Out Q8, section 5.2 p.114 on twin primes if not yet completed.

Period 5 · 40 min · Safe Treasures: Recognising Co-prime Pairs

Material	Textbook pages 115-116, Board space for factor lists, Pair recording slips — **Pedagogical approach:** Inductive
0-8	Recall the Jump Jackpot game from section 5.1 and reintroduce it with the new rule: jump size 1 is forbidden. Ask students: if treasures are on 15 and 39, is Grumpy safe? Students list factors of 15 and 39 to find the answer, inducing the idea that 'safe' pairs share no common factor greater than 1.
8-22	Define co-prime pairs formally and work through Figure it Out Q1, section 5.3 p.115 ("check if (15,39), (4,15), (18,29), (20,55) are co-prime") together, then students solve Figure it Out Q2, section 5.3 p.116 ("which of five pairs are co-prime?") individually. Circulate and note which students list all factors versus only a few.
22-33	Run the Math Talk from Math Talk, section 5.3 p.116 ("when does LCM equal the product vs. less than the product?"). Students generate their own examples in pairs and share; teacher records co-prime pairs and non-co-prime pairs on the board to build the pattern inductively.
33-40	Consolidate: students state in their own words the definition of co-prime and give one example of a co-prime pair and one non-co-prime pair that surprised them. Preview the Co-prime Art activity coming in the next period.
Teacher Notes	Transition: connect the 'no common factor > 1' idea back to the Jump Jackpot game so students see co-primality as a natural extension of Period 2's work. A common error is assuming co-prime numbers must both be prime; use the pair (4, 15) from Q1 to show otherwise. Figure it Out Q2(d), section 5.3 p.116 (17 and 69) is a useful self-study check since 17 is prime but 69 = 3 x 23.

Period 6 · 60 min · Co-prime Art and the Need for Prime Factorisation

Material	Textbook pages 116-120, Dot-grid paper for thread-art diagrams, Factor tree templates — **Pedagogical approach:** Discovery
0-15	Students work through Activity (Co-prime Art), section 5.3 p.116 ("predict which peg/gap combinations visit all pegs vs. only some...."). They draw patterns for at least two of the four specified combinations and verify their co-primality predictions using factor lists from Period 5.
15-25	Transition to section 5.4 via the Anshu-Guna dialogue: 56 = 14 x 4 hides the factor 7 that 56 and 63 share. Pose the question: how do we find ALL factors reliably? Introduce prime factorisation as the systematic solution and show the tree diagram for 36 yielding 2^2 x 3^2 regardless of factorisation path.
25-42	Students work individually on a selected subset of Figure it Out Q1, section 5.4 p.120 ("find prime factorisations of 64, 104, 105, 243, 320..."), using factor trees. Circulate to ensure students break down to primes fully and write the repeated-prime notation correctly.
42-52	Work through Figure it Out Q2, section 5.4 p.120 ("a number has one 2, two 3s, one 11 in its factorisation; what is it?") as a class to reinforce reading factorisation in reverse. Briefly state the uniqueness of prime factorisation as a remarkable fact.
52-60	Assign Figure it Out Q4, section 5.4 p.120 as homework and preview next period's applications: using prime factorisation to check co-primality and divisibility without guessing.
Homework	Figure it Out Q4, section 5.4 p.120 — Find the prime factorisation without multiplying first: (a) 56 × 25; (b) 108 × 75;…
Teacher Notes	Transition: the Co-prime Art activity consolidates section 5.3's apply goal, and the Anshu-Guna dialogue immediately motivates section 5.4 - keep the two halves of this period tightly linked. Watch for students stopping factor trees at composite numbers (e.g., writing 36 = 4 x 9 and stopping); insist on continuing to primes. Example, section 5.4 p.120 (WE-1) on checking co-primality of 40 and 231 is a useful self-study pointer.

Period 7	40 min	Prime Factorisation as a Reasoning Tool: Co-primality and Divisibility
Material	Textbook pages 120-122, Student factor-tree work from Period 6	**Pedagogical approach:** Deductive
0-8	Review homework Figure it Out Q4, section 5.4 p.120 and check that students can combine prime factorisations of two factors without re-multiplying. Then walk through Example, section 5.4 p.121 (WE-3) ("168 = 2^3x3x7 is divisible by 12 = 2^2x3 because 12's factorisation is contained in 168's") as a deductive template.	
8-22	Students work on Figure it Out Q1, section 5.4 p.122 ("guess then verify co-primality for (30,45), (57,85), (121,1331), (343,216)") using the deductive method: factorise both, compare prime lists. Emphasise that absence of any shared prime guarantees co-primality.	
22-33	Work through Figure it Out Q2, section 5.4 p.122 ("is first number divisible by second? use prime factorisation") and Figure it Out Q3, section 5.4 p.122 ("two numbers given as factorisations; are they co-prime? does one divide the other?") as paired exercises, reinforcing the logical equivalence between divisibility and factorisation containment.	
33-40	Discuss Figure it Out Q4, section 5.4 p.122 ("are any two primes always co-prime?") as a class reasoning question. Consolidate: students write the deductive rule for co-primality check and preview divisibility tests by short-cut rules coming next.	
Teacher Notes	Transition: connect Period 6's factor-tree fluency to today's deductive application - the tree is a tool; today's goal is to reason with it. A common error is concluding divisibility from only one shared prime factor; use Example, section 5.4 p.122 (WE-5) ("42 not divisible by 12 because 2 appears only once in 42") to address this. Example, section 5.4 p.121 (WE-2) on 242 and 195 may serve as a self-study pointer.	

Period 8	50 min	Last-Digit Logic: Deriving Divisibility Tests for 2, 4, 5, 8, and 10
Material	Textbook pages 123-126, Divisibility rule summary table (blank template), Calculators optional for verification	**Pedagogical approach:** Inductive
0-10	Open with the motivation from section 5.5: finding factors of large numbers by trial division is tedious. Ask students to list all multiples of 10 up to 100, then all multiples of 5 - what do they notice about the last digit? Inductively derive the rules for 10 and 5.	
10-22	Extend the pattern to 2 (even last digit). Then pose the challenge: 12 and 22 have the same units digit, but only 12 is divisible by 4 - a single digit is insufficient. Guide students to examine multiples of 4 and discover through exploration that only the last two digits determine divisibility by 4.	
22-35	Present the three-statement equivalence for divisibility by 4 from the textbook and ask students to verify each using examples from Figure it Out Q3, section 5.5 p.125 ("always/sometimes/never: sum of two evens is a multiple of 4"). Then extend the reasoning pattern to divisibility by 8 using the last three digits.	
35-45	Students work on Figure it Out Q4, section 5.5 p.126 ("find remainders when dividing by 10, 5, 2") and Figure it Out Q5, section 5.5 p.126 ("which two checks suffice to certify divisibility by all five?"). Discuss the logical reasoning behind the Q5 answer as a class.	
45-50	Summarise all five divisibility rules as a class-constructed table. Note that tests for 3, 6, 7, 9 come in later classes. Assign Figure it Out Q6, section 5.5 p.126 as homework.	
Homework	Figure it Out Q6, section 5.5 p.126 — Which of 572, 2352, 5600, 6000, 77622160 are divisible by all of 2, 4, 5,…	
Teacher Notes	Transition: link the divisibility tests back to prime factorisation from Periods 6-7 - the tests are short-cuts that avoid full factorisation. Watch for students over-generalising the last-two-digit rule to divisibility by 6 or other numbers; clarify that this particular structure arises from 100 being divisible by 4. Figure it Out Q1, section 5.5 p.125 on leap years is a rich self-study problem combining divisibility by 4, 100, and 400.	

Period 9	50 min	Applying Divisibility Tests: Leap Years, Palindromes, and Clever Checks
Material	Textbook pages 125-126, Completed divisibility rules table from Period 8	**Pedagogical approach:** Problem-solving
0-12	Review homework Figure it Out Q6, section 5.5 p.126 from Period 8. Then launch Figure it Out Q1, section 5.5 p.125 ("2024 is a leap year; identify leap years from birth year to now and count from 2024 to 2099"). Students apply the divisibility by 4, 100, and 400 rules in combination - a multi-condition problem.	

12-27	Students work on Figure it Out Q2, section 5.5 p.125 ("find largest and smallest 4-digit palindromes divisible by 4"). Pairs discuss strategy: what constraints does a palindrome place on the last two digits? Guide students to see that divisibility by 4 depends only on those last two digits.
27-40	Work through Figure it Out Q7, section 5.5 p.126 ("two numbers whose product is 10000, neither ending in 0"). Students use prime factorisation ($10000 = 2^4 \times 5^4$) to split factors creatively, applying both factorisation and divisibility knowledge together.
40-50	Class discussion: which problem in this section required the most steps of reasoning? Students share strategies and the teacher consolidates key ideas. Preview section 5.6 as a playful application chapter where number-theory knowledge is used in open-ended puzzles.
Teacher Notes	Transition: note that Period 8 built the divisibility rules deductively; today applies them to novel, multi-condition problems. A common error in the palindrome problem is forgetting that a 4-digit palindrome has the form abba, so the last two digits are 'ba' - remind students to check this form explicitly. Figure it Out Q3, section 5.5 p.125 on always/sometimes/never is a good self-study extension for fast finishers.

Period 10	40 min	**Special Numbers and the Prime Puzzle: Creative Reasoning with Number Theory**

Material	Textbook pages 126-127, Grid worksheets (optional, for larger Prime Puzzle), Board space for recording special-number reasons	**Pedagogical approach:** Problem-solving
0-15	Launch Math Talk (Special Numbers), section 5.6 p.126 ("explain what makes each number special compared to the others in its box...."). Students work individually for 3 minutes, then share in small groups; the teacher records diverse valid reasons on the board, emphasising that the same number can be special for multiple reasons.	
15-33	Introduce Math Talk (Prime Puzzle), section 5.6 p.127 ("fill grid with primes; row products and column products given...."). Students begin with sub-part (a), the smaller grid. Encourage them to use prime factorisation of the given row/column products as the entry strategy.	
33-40	Pairs who complete the first grid attempt sub-part (b), the larger grid. Close the period with a chapter-level consolidation: the teacher asks students to name one concept from each section (multiples/factors, primes/sieve, co-primes, prime factorisation, divisibility tests) in a rapid-fire oral round.	
Teacher Notes	Transition: frame this period as 'bringing it all together' - every concept from the chapter appears in the Special Numbers boxes or the Prime Puzzle. Resist the urge to provide a unique correct answer for the Special Numbers activity; productive disagreement is the goal. The Prime Puzzle (section 5.6 p.127) is the richest self-study item for students who want an additional challenge.	